

A Queen Mole-Rat's Scent Helps Keep Her Colony Calm

From: Science News Explores Mined: 2026-10-03 3 min read

Naked mole-rats live together in groups called **colonies**. Some members search for food. Others care for the young. The queen has babies. A study found that a special **scent** from the queen helps keep other members from fighting or **reproducing**, which means having young.

Gary Lewin studies how naked mole-rats use their senses. He is a scientist at the Max Delbrück Center in Berlin, Germany. He had found that these animals use **scent** to recognize mole-rats from other **colonies**. His team wanted to learn whether smells also matter within a colony.

The researchers found a **chemical** called isopropyl myristate that only the queens produced among the mole-rats they studied. It activated the part of the animals' brains that handles smells.

To test the **chemical**, the team removed a queen from her colony. They then added the **chemical** every day for three months. During that time, the animals did not fight or mate. Mating is part of animal reproduction. When the team stopped adding the **scent**, fighting began. A few weeks later, the animals began mating, and a new queen emerged.

In another experiment, the researchers separated male and female pairs from the colony. Some pairs received the queen's **scent**; others did not. After four weeks, none of the females exposed to the **scent** had mated. Five of the six females without it were **pregnant**.

The team also found that the **chemical** changed **hormones** in female mole-rats. **Hormones** are **chemical** signals in the body. These findings help explain how a queen's **scent** can affect life in a colony.

Word Treasure

colonies -- Groups of animals that live together.

scent -- A smell.

reproducing -- Having young.

chemical -- A substance with a particular makeup.

pregnant -- Having developing young inside the body.

hormones -- Chemical signals in the body.

Background you need

A colony is a group of animals living together. Naked mole-rats in this article share tasks such as finding food and caring for young.

Reproduction means having young. Hormones are chemical signals in the body.

Break it down

Introduces colony teamwork and the queen's scent.

Explains Lewin's question and the chemical his team found.

Describes the colony experiment and what happened when the scent stopped.

Describes the pair experiment and introduces hormone changes.

Why it matters

The experiments help explain how a queen's scent affects both behavior and reproduction in a naked mole-rat colony.

Test what you remember

1. What work do some naked mole-rats do in their colony?

- ☐ (A) Measure their hormones.
- ☐ (B) Add the test chemical.
- ☐ (C) Study their brains.
- ☐ (D) Search for food.

2. What did Lewin's team want to learn about smells?

- ☐ (A) Whether they change the food.
- ☐ (B) Whether they reach other cities.
- ☐ (C) Whether they matter within a colony.
- ☐ (D) Whether they come from the young.

3. Which part of the body did the chemical activate?

- ☐ (A) The brain part that handles sounds.
- ☐ (B) The brain part that handles smells.
- ☐ (C) The part that digests the food.
- ☐ (D) The part that moves the legs.

4. How long did researchers add the chemical every day?

- ☐ (A) For six months.
- ☐ (B) For one week.
- ☐ (C) For three months.
- ☐ (D) For four weeks.

5. What began when researchers stopped adding the scent?

- ☐ (A) Testing by a new team.
- ☐ (B) Feeding by the researchers.
- ☐ (C) Care for a new colony.
- ☐ (D) Fighting among the mole-rats.

6. What happened to five of the six females without the scent in the pair experiment?

- ☐ (A) They received the scent.
- ☐ (B) They were pregnant.
- ☐ (C) They became new queens.
- ☐ (D) They joined another colony.

Answer key (try the quiz first!): 1: D 2: C 3: B 4: C 5: D 6: B

MY THOUGHTS (at least 20 words):